

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	A16
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Explainable AI for Autonomous Anomaly Detection in Hybrid Cloud Data Infrastructure

Abstract

The rapid adoption of hybrid cloud computing has transformed the way organizations store, process, and manage data by integrating private and public cloud infrastructures. However, the increasing complexity of hybrid cloud environments has introduced significant challenges in ensuring system security, reliability, and operational efficiency. Organizations continuously generate massive volumes of network traffic, system logs, application events, and performance metrics, making it difficult for administrators to manually identify abnormal activities. Cyberattacks, unauthorized access, system failures, resource misuse, and performance degradation can occur unexpectedly, causing financial losses, service disruptions, and security breaches. Existing anomaly detection systems primarily rely on rule-based methods or black-box machine learning models that detect anomalies without providing clear explanations for their decisions. As a result, administrators often struggle to understand why an anomaly was detected, reducing trust in AI-based security systems and delaying incident response.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), Explainable Artificial Intelligence (XAI), and cloud security have demonstrated the potential of intelligent systems to improve anomaly detection in cloud infrastructures. However, most existing solutions focus mainly on achieving high detection accuracy while providing limited transparency and interpretability. Furthermore, many systems require continuous human intervention for monitoring and decision-making rather than offering autonomous detection and explanation capabilities within a unified framework.

This project proposes **Explainable AI for Autonomous Anomaly Detection in Hybrid Cloud Data Infrastructure**, an intelligent system that combines Machine Learning and Explainable Artificial Intelligence (XAI) to detect and explain anomalies in hybrid cloud environments. The system continuously monitors cloud data, identifies abnormal patterns such as cyber threats, system failures, and performance issues, and provides human-readable explanations using XAI techniques. It also generates real-time alerts to support faster and more reliable decision-making.

The proposed system focuses on autonomous anomaly detection, continuous cloud monitoring, explainable decision-making, and intelligent threat detection. It aims to improve detection accuracy, reduce false alarms, enhance transparency, and increase trust in AI-driven cloud security systems. By integrating Machine Learning and Explainable AI, the system provides a scalable and research-oriented solution for securing modern hybrid cloud data infrastructures.

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References (Category specific)

Base Paper

1. Alam, K., et al., "SXAD: Shapely eXplainable AI-Based Anomaly Detection Using Log Data." IEEE Access, 2024

Link: <https://doi.org/10.1109/ACCESS.2024.3425472>

Supporting Papers

2. Ahmed, U., et al., "Explainable AI-Based Innovative Hybrid Ensemble Model for Intrusion Detection." Journal of Cloud Computing, 2024.

Link: <https://doi.org/10.1186/s13677-024-00712-x>

3. Gaspar, D., et al., "Explainable AI for Intrusion Detection Systems: LIME and SHAP Applicability on Multi-Layer Perceptron." IEEE Access, 2024

Link: <https://doi.org/10.1109/ACCESS.2024.3368377>

4. Nazat, S., et al., "On Evaluating Black-Box Explainable AI Methods for Enhancing Anomaly Detection." Sensors, 2024

Link: <https://doi.org/10.3390/s24113515>

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